

Figure 1 A Computer Network.

**Remark:** The set of vertices  $V$  of a graph  $G$  may be infinite. A graph with an infinite vertex set is called an **infinite graph**, and in comparison, a graph with a finite vertex set is called a **finite graph**. In this book we will usually consider only **finite graphs**.

Now suppose that a network is made up of data centers and communication links between computers. We can represent the location of each data center by a point and each communications link by a line segment, as shown in Figure 1.

This **computer network** can be modeled using a graph in which the **vertices** of the graph represent the **data centers** and the **edges** represent **communication links**. In general, we visualize graphs by using points to represent vertices and line segments, possibly curved, to represent edges, where the endpoints of a line segment representing an edge are the points representing the endpoints of the edge. When we draw a graph, we generally **try to draw edges** so that **they do not cross**. However, this is not necessary because any depiction using points to represent vertices and any form of connection between vertices can be used. Indeed, there are some graphs that cannot be drawn in the plane without edges crossing (see Section 8.7). The key point is that the way we draw a graph is arbitrary, as long as the correct connections between vertices are depicted.

Note that **each edge** of the graph representing this **computer network** connects **two different vertices**. That is, **no edge connects a vertex to itself**. Furthermore, **no two different edges connect the same pair of vertices**. A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a **simple graph**. Note that in a simple graph, each edge is associated to an unordered pair of vertices, and no other edge is associated to this same edge. Consequently, when there is an edge of a simple graph associated to  $\{u, v\}$ , we can also say, without possible confusion, that  $\{u, v\}$  is an edge of the graph.

A **computer network** may contain **multiple links** between data centers, as shown in Figure 2. To model such networks we need graphs that have more than one edge connecting the same pair of vertices. Graphs that may have **multiple edges** connecting the same vertices are called **multigraphs**. When there are  $m$  different edges associated to the same unordered pair of vertices  $\{u, v\}$ , we also say that  $\{u, v\}$  is an **edge of multiplicity  $m$** . That is, we can think of this set of edges as  $m$  different copies of an edge  $\{u, v\}$ .

Sometimes a **communications link connects a data-center with itself**, perhaps a **feedback loop for diagnostic purposes**. Such a network is illustrated in Figure 3. To model this network we need to include edges that connect a vertex to itself. Such edges are called **loops**. Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices, are sometimes called **pseudographs**.

So far the graphs we have introduced are **undirected graphs**. Their edges are also said to be **undirected**. However, to construct a graph model, we may find it necessary to assign directions to the edges of a graph.

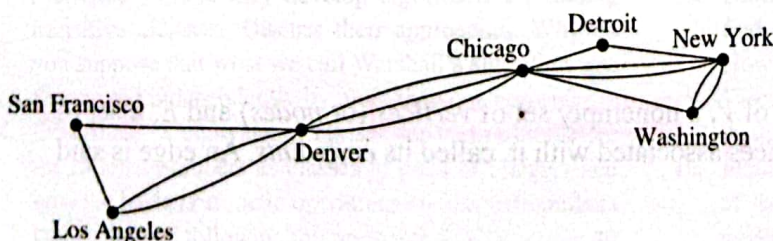


Figure 2 A Computer Network with Multiple Links between Data Centers.

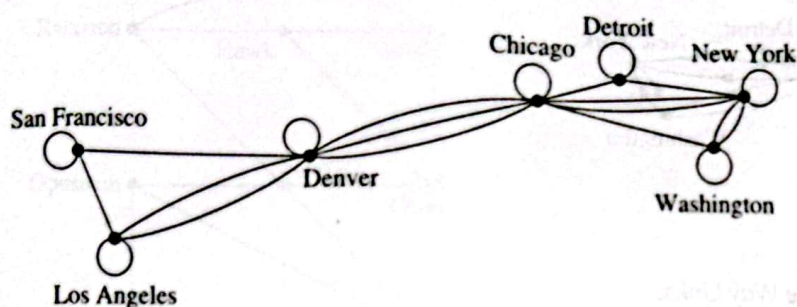


Figure 3 A Computer Network with Diagnostic Links.

For example, in a computer network, some links may operate in only one direction (such links are called single duplex lines). This may be the case if there is a large amount of traffic sent to some data centers, with little or no traffic going in the opposite direction. Such a network is shown in Figure 4.

To model such a computer network we use a directed graph. Each edge of a directed graph is associated to an ordered pair. The definition of directed graph we give here is more general than the one we used in Chapter 7, where we used directed graphs to represent relations.

**Definition 2** A **directed graph** (or **digraph**)  $(V, E)$  consists of a **nonempty set of vertices**  $V$  and a **set of directed edges** (or **arcs**)  $E$ . Each **directed edge** is associated with an **ordered pair of vertices**. The directed edge associated with the ordered pair  $(u, v)$  is said to **start at  $u$**  and **end at  $v$** .

When we depict a directed graph with a line drawing, we use an arrow pointing from  $u$  to  $v$  to indicate the direction of an edge that starts at  $u$  and ends at  $v$ . A directed graph may contain loops and it may contain multiple directed edges that start and end at the same vertices. A directed graph may also contain directed edges that connect vertices  $u$  and  $v$  in both directions; that is, when a digraph contains an edge from  $u$  to  $v$ , it may also contain one or more edges from  $v$  to  $u$ . Note that we obtain a directed graph when we assign a direction to each edge in an undirected graph. When a directed graph has **no loops and has no multiple directed edges**, it is called a **simple directed graph**. Because a simple directed graph has at most one edge associated to each ordered pair of vertices  $(u, v)$ , we call  $(u, v)$  an edge if there is an edge associated to it in the graph.

In some computer networks, multiple communication links between two data centers may be present, as illustrated in Figure 5. Directed graphs that may have multiple directed edges from a vertex to a second (possibly the same) vertex are used to model such networks. We called such graphs **directed multigraphs**. When there are  $m$  directed edges, each associated to an ordered pair of vertices  $(u, v)$ , we say that  $(u, v)$  is an edge of **multiplicity  $m$** .

For some models we may need a graph where some edges are undirected, while others are directed. A graph with both directed and undirected edges is called a **mixed graph**. For example, a mixed graph might be used to model a computer network containing links that operate in both directions and other links that operate only in one direction.

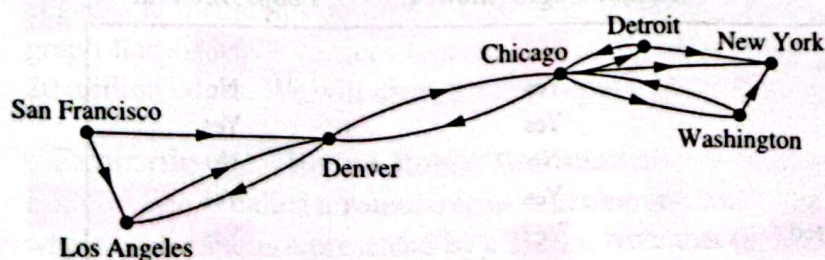
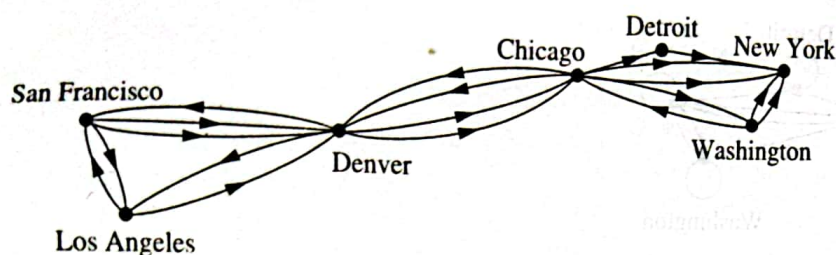


Figure 4 A Communications Network with One-Way Communications Links.



**Figure 5** A Computer Network with Multiple One-Way Links.

This terminology for the various types of graphs is summarized in Table 1. We will sometimes use the term **graph** as a general term to describe graphs with directed or undirected edges (or both), with or without loops, and with or without multiple edges. At other times, when the context is clear, we will use the term **graph** to refer only to undirected graphs.



Because of the relatively modern interest in graph theory, and because it has applications to a wide variety of disciplines, many different terminologies of graph theory have been introduced. The reader should determine how such terms are being used whenever they are encountered.

The terminology used by mathematicians to describe graphs has been increasingly standardized, but the terminology used to discuss graphs when they are used in other disciplines is still quite varied.

Although the terminology used to describe graphs may vary, three key questions can help us understand the structure of a graph:

- Are the edges of the graph undirected or directed (or both)?
- If the graph is undirected, are multiple edges present that connect the same pair of vertices? If the graph is directed, are multiple directed edges present?
- Are loops present?

Answering such questions helps us understand graphs. It is less important to remember the particular terminology used.

### Graph Models

Graphs are used in a wide variety of models. We will present a few graph models from diverse fields here. Others will be introduced in subsequent sections of this and later chapters.



When we build a graph model, we need to make sure that we have correctly answered the three key questions we posed about the structure of a graph.

### Example 1

#### Niche Overlap Graphs in Ecology

Graphs are used in many models involving the interaction of different species of animals. For instance, the competition between species in an ecosystem can be modeled using a **niche overlap graph**. Each species is represented by



**Table 1** Graph Terminology.

| Type                  | Edges                   | Multiple Edges Allowed? | Loops Allowed? |
|-----------------------|-------------------------|-------------------------|----------------|
| Simple graph          | Undirected              | No                      | No             |
| Multigraph            | Undirected              | Yes                     | No             |
| Pseudograph           | Undirected              | Yes                     | Yes            |
| Simple directed graph | Directed                | No                      | No             |
| Directed multigraph   | Directed                | Yes                     | Yes            |
| Mixed graph           | Directed and undirected | Yes                     | Yes            |

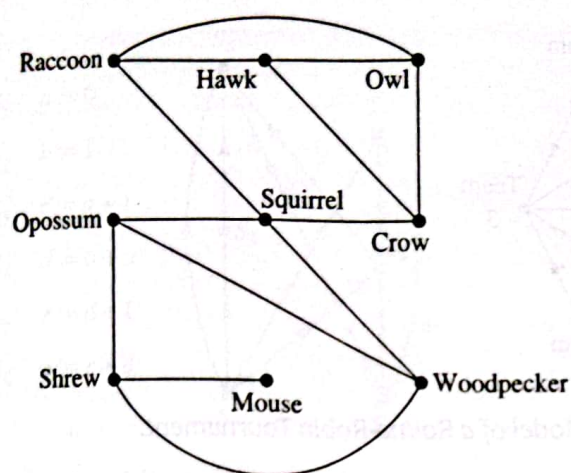


Figure 6 A Niche Overlap Graph

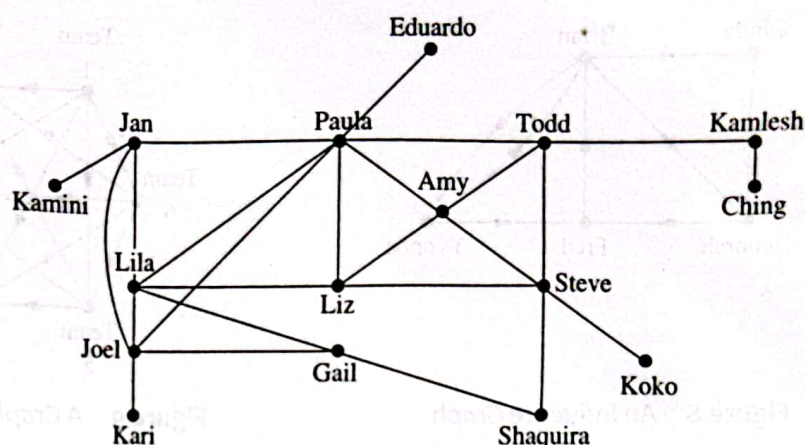


Figure 7 An Acquaintanceship Graph.

a vertex. An undirected edge connects two vertices if the two species represented by these vertices compete (that is, some of the food resources they use are the same). A **niche overlap graph** is a **simple graph** because **no loops or multiple edges** are needed in this model. The graph in Figure 6 models the ecosystem of a forest. We see from this graph that squirrels and raccoons compete but that crows and shrews do not. ◀

## Example 2 Acquaintanceship Graphs

**Links** We can use graph models to **represent various relationships between people**. For example, we can use a **simple graph** to represent **whether two people know each other**, that is, whether they are acquainted. Each person in a particular group of people is represented by a vertex. An **undirected edge** is used to connect two people when these people know each other. **No multiple edges** and usually **no loops** are used. (If we want to include the notion of **self-knowledge**, we would **include loops**.) A small acquaintanceship graph is shown in Figure 7. The acquaintanceship graph of all people in the world has **more than six billion vertices** and probably **more than one trillion edges**! We will discuss this graph further in Section 8.4. ◀

**Example 3 Influence Graphs** In **studies of group behavior**, it is observed that **certain people can influence the thinking of others**. A **directed graph** called an **influence graph** can be used to model this behavior. Each person of the group is represented by a vertex. There is a directed edge from vertex  $a$  to vertex  $b$  when the person represented by **vertex  $a$  influences the person represented by vertex  $b$** . This graph **does not contain loops** and it **does not contain multiple directed edges**. An example of an influence graph for members of a group is shown in Figure 8. In the group modeled by this influence graph, **Deborah can influence Brian, Fred, and Linda**, but **no one can influence her**. Also, **Yvonne and Brian can influence each other**. ◀

**Example 4 The Hollywood Graph** The Hollywood graph represents **actors by vertices** and connects two vertices when the actors represented by **these vertices have worked together on a movie**. This graph is a **simple graph** because its edges are **undirected**, it contains **no multiple edges**, and it contains **no loops**. According to the Internet Movie Database, in January 2006 the Hollywood graph had 637,099 vertices representing actors who have appeared in 339,896 films, and had more than 20 million edges. We will discuss some aspects of the Hollywood graph later in Section 8.4. ◀

**Example 5 Round-Robin Tournaments** A tournament where **each team plays each other team exactly once** is called a **round-robin tournament**. Such tournaments can be modeled using **directed graphs** where each team is represented by a vertex. Note that  **$(a, b)$  is an edge if team  $a$  beats team  $b$** . This graph is a

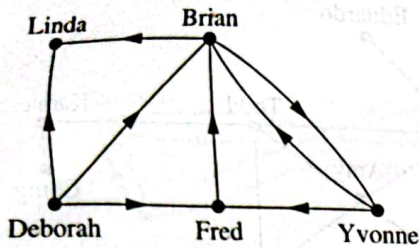


Figure 8 An Influence Graph.

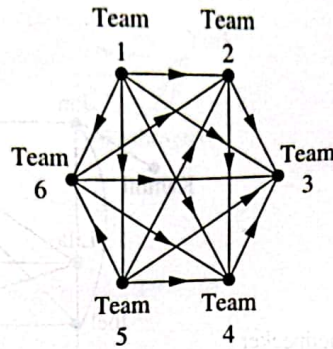


Figure 9 A Graph Model of a Round-Robin Tournament.

simple directed graph, containing no loops or multiple directed edges (because no two teams play each other more than once). Such a directed graph model is presented in Figure 9. We see that Team 1 is undefeated in this tournament, and Team 3 is winless.

**Example 6 Collaboration Graphs** A graph called a collaboration graph can be used to model joint authorship of academic papers. In a collaboration graph, vertices represent people (perhaps restricted to members of a certain academic community) and edges link two people if they have jointly written a paper. This graph is a simple graph because it contains undirected edges and has no loops or multiple edges. The collaboration graph for people working together on research papers in mathematics has been found to have more than 400,000 vertices and 675,000 edges. We will have more to say about this graph in Section 8.4.

**Example 7 Call Graphs** Graphs can be used to model telephone calls made in a network, such as a long-distance telephone network. In particular, a directed multigraph can be used to model calls where each telephone number is represented by a vertex and each telephone call is represented by a directed edge. The edge representing a call starts at the telephone number from which the call was made and ends at the telephone number to which the call was made. We need directed edges because the direction in which the call is made matters. We need multiple directed edges because we want to represent each call made from a particular telephone number to a second number.

A small telephone call graph is displayed in Figure 10(a), representing seven telephone numbers. This graph shows, for instance, that three calls have been made from 732-555-1234 to 732-555-9876 and two in the other direction, but no calls have been made from 732-555-4444 to any of the other six numbers except 732-555-0011. When we care only whether there has been a call connecting two telephone numbers, we use

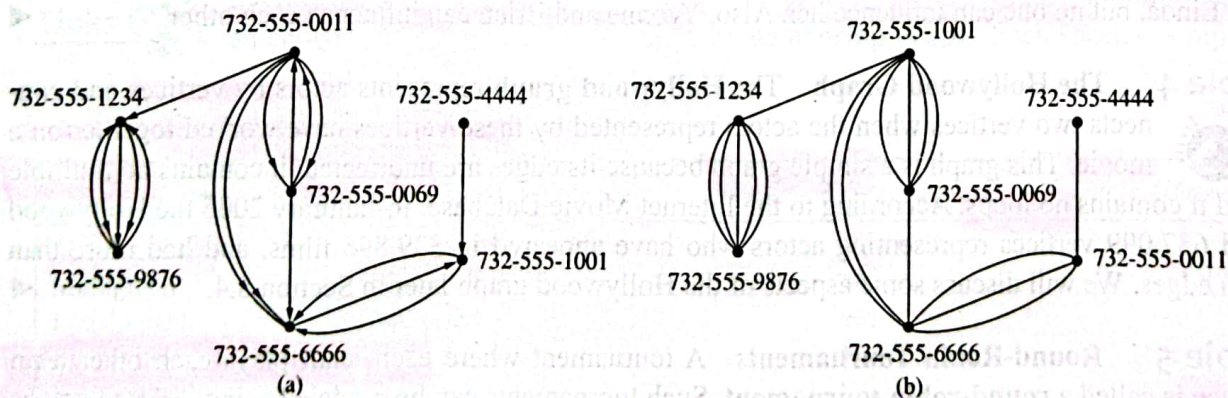


Figure 10 A Call Graph.

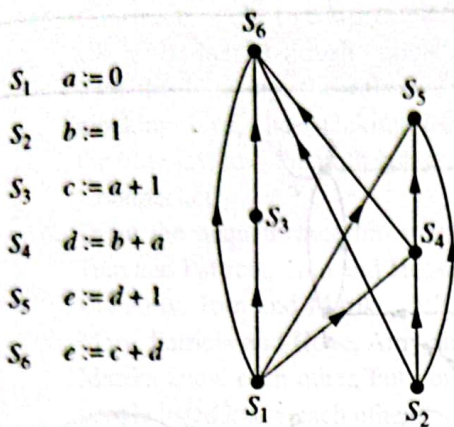


Figure 11 A Precedence Graph.

an undirected graph with an edge connecting telephone numbers when there has been a call between these numbers. This version of the call graph is displayed in Figure 10(b).

Call graphs that model actual calling activities can be huge. For example, one call graph studied at AT&T, which models calls during 20 days, has about 290 million vertices and 4 billion edges. We will discuss call graphs further in Section 8.4.

**Example 8 The Web Graph** The World Wide Web can be modeled as a directed graph where each Web page is represented by a vertex and where an edge starts at the Web page  $a$  and ends at the Web page  $b$  if there is a link on  $a$  pointing to  $b$ . Because new Web pages are created and others removed somewhere on the Web almost every second, the Web graph changes on an almost continual basis. Currently the Web graph has more than three billion vertices and 20 billion edges. Many people are studying the properties of the Web graph to better understand the nature of the Web. We will return to Web graphs in Section 8.4, and in Chapter 9 we will explain how the Web graph is used by the Web crawlers that search engines use to create indices of Web pages.

**Example 9 Precedence Graphs and Concurrent Processing** Computer programs can be executed more rapidly by executing certain statements concurrently. It is important not to execute a statement that requires results of statements not yet executed. The dependence of statements on previous statements can be represented by a directed graph. Each statement is represented by a vertex, and there is an edge from one vertex to a second vertex if the statement represented by the second vertex cannot be executed before the statement represented by the first vertex has been executed. This graph is called a precedence graph. A computer program and its graph are displayed in Figure 11. For instance, the graph shows that statement  $S_5$  cannot be executed before statements  $S_1$ ,  $S_2$ , and  $S_4$  are executed.

**Example 10 Roadmaps** Graphs can be used to model roadmaps. In such models, vertices represent intersections and edges represent roads. Undirected edges represent two-way roads and directed edges represent one-way roads. Multiple undirected edges represent multiple two-way roads connecting the same two intersections. Multiple directed edges represent multiple one-way roads that start at one intersection and end at a second intersection. Loops represent loop roads. Consequently, roadmaps depicting only two-way roads and no loop roads, and in which no two roads connect the same pair of intersections, can be represented using a simple undirected graph. Roadmaps depicting only one-way roads and no loop roads, and where no two roads start at the same intersection and end at the same intersection, can be modeled using simple directed graphs. Mixed graphs are needed to depict roadmaps that include both one-way and two-way roads.